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Deep Learning for Student Engagement Detection: A Lightweight CNN Approach for Educational Environments

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Abstract

Automated student engagement detection from facial expressions is a promising approach for real-time monitoring in educational environments. This paper proposes a lightweight deep learning framework integrating MobileNetV3 and EfficientNet-B0 with a Channel-Spatial Attention Module (CSAM) and focal weighted loss for engagement recognition. CSAM is a purpose-built dual-attention module optimized for mobile architectures, featuring compact channel attention (reduction ratio $r = 4$) and per-block feature recalibration that jointly directs model focus toward engagement-relevant facial regions. We map the FER2013 facial expression dataset (35,887 images) and CK+ dataset (804 images) to a four-class engagement taxonomy (Engaged, Boredom, Frustration, Confusion) via an emotion-to-engagement mapping grounded in affective computing literature. Our training pipeline combines two-stage transfer learning from ImageNet, data augmentation, focal weighted loss with label smoothing, and weighted sampling for joint class balancing and hard-example mining. MobileNetV3+CSAM achieves 70.5% accuracy on the four-class task with 1.26M parameters and 0.068G FLOPs—a 19.6$\times$ energy reduction compared to VGG-16. EfficientNet-B0+CSAM reaches 69.1% with 5.4M parameters. CSAM outperforms existing attention mechanisms (SE-Net, ECA-Net, CBAM) by 2.4–3.7% on the same backbone. Ablation studies confirm the contributions of data augmentation (+3.8%), focal weighted loss (+1.5–2.4% F1 on minority classes), transfer learning (+9.2%), and focal loss parameter selection ($\gamma = 1.0$ outperforms $\gamma = 2.0$ by 0.9% F1). Cross-dataset evaluation on CK+ (71.9% accuracy without fine-tuning, 114 test samples) provides preliminary evidence of generalizability. Validation on DAiSEE—an engagement-specific dataset with direct annotations—achieves 74.6% validation accuracy and 61.2% test accuracy on binary engagement classification (despite extreme class imbalance), providing preliminary evidence that the framework generalizes beyond emotion-to-engagement mapping. The novel combination of CSAM, focal weighted loss, emotion-to-engagement mapping, and deployment-oriented evaluation constitutes a coherent framework that bridges the gap between engagement recognition accuracy and real-world deployability.

Keywords: student engagement; facial expression recognition; MobileNetV3; lightweight CNN; attention mechanism

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1. Introduction

Student engagement is one of the strongest predictors of academic success and learning outcomes. Engaged students show greater attention, participation, and persistence, all of which support deeper learning (Dixson, 2015; Fredricks et al., 2004). Yet monitoring

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engagement in real time remains a persistent challenge for educators, particularly in large classes and online settings where manual observation is impractical (Whitehill et al., 2014). The expansion of online and blended education has further increased the need for automated, scalable engagement monitoring tools (Mandia et al., 2024).

Computer vision offers a promising avenue: facial expressions encode cognitive and emotional states, including interest, confusion, boredom, and frustration, that correlate with engagement levels (Jampour & Javidi, 2022; Kleinsmith & Bianchi-Berthouze, 2013). Convolutional neural networks (CNNs) have advanced facial expression recognition substantially (Bengio et al., 2013), yet deploying them in real classroom environments faces practical computational constraints. Models such as ResNet-50 and VGG-16 demand memory and processing power that exceed the capacity of edge devices commonly available in schools (A. G. Howard et al., 2017). This gap between recognition accuracy and deployability on resource-limited hardware motivates the search for efficient architectures suitable for classroom use.

While prior work has demonstrated the feasibility of CNN-based engagement detection (Mandia et al., 2024; Tonguç & Ozkara, 2020; Whitehill et al., 2014), three gaps remain. First, most existing methods employ heavyweight architectures (ResNet-50, VGG-16) that are impractical for real-time classroom deployment on edge devices. Second, attention mechanisms designed for large backbones (e.g., CBAM (Woo et al., 2018)) have not been specifically adapted for mobile-class architectures used in engagement detection. Third, class imbalance in engagement datasets—where disengaged states are underrepresented—is typically addressed with simple class weighting rather than joint hard-example mining and balanced sampling.

To address these gaps, this paper investigates lightweight deep learning architectures for student engagement detection. Specifically, we evaluate MobileNetV3 (A. Howard et al., 2019) and EfficientNet-B0 (Tan & Le, 2019), which are designed to achieve competitive accuracy while maintaining low computational overhead. We repurpose the FER2013 facial expression dataset (Goodfellow et al., 2013) and CK+ (Lucey et al., 2010) for engagement detection through an emotion-to-engagement mapping, converting the seven basic emotions into four engagement states (Engaged, Boredom, Frustration, Confusion).

The main contributions of this paper are as follows:

1. We propose a lightweight CNN framework integrated with a Channel-Spatial Attention Module (CSAM), a purpose-built dual-attention module for mobile architectures with compact channel attention (reduction ratio $r = 4$) and per-block feature recalibration. CSAM outperforms existing attention mechanisms (SE-Net, ECA-Net, CBAM) by 2.4–3.7% on the same backbone, with only 0.3% parameter overhead.
2. We introduce a focal weighted loss function with label smoothing and gradient clipping that jointly addresses class imbalance and hard-example mining, outperforming standard weighted cross-entropy by 1.5% F1 on Boredom and 2.4% F1 on Confusion. We further employ weighted sampling to ensure balanced class representation during training.
3. We repurpose FER2013 and CK+ for engagement detection through an emotion-to-engagement mapping grounded in affective computing theory, enabling evaluation on publicly available datasets with well-established benchmarks.
4. We conduct comprehensive deployment-oriented experiments comparing lightweight models with heavier architectures, demonstrating that MobileNetV3+CSAM achieves 70.5% accuracy with only 1.26M parameters and 0.068G FLOPs, offering an optimal accuracy-efficiency trade-off.

The remainder of this paper is organized as follows. Section 2 reviews related work on engagement detection and lightweight CNN architectures. Section 3 describes the proposed methodology and presents the experimental results. Section 4 discusses the findings and limitations. Section 5 concludes the paper.

2. Related Work

2.1. Student Engagement Detection

Student engagement is a multidimensional construct encompassing behavioral, emotional, and cognitive components (Fredricks et al., 2004). Behavioral engagement refers to participation and effort, emotional engagement encompasses affective reactions such as interest, boredom, and frustration, and cognitive engagement involves self-regulation and investment in learning (Skinner et al., 2009). Early approaches to automatic engagement detection relied on behavioral cues such as gaze direction, head pose, and body posture (Whitehill et al., 2014). Whitehill et al. (Whitehill et al., 2014) pioneered the use of facial expressions for engagement recognition, demonstrating that computer vision systems can achieve reasonable accuracy in distinguishing engaged from disengaged states.

CNN-based approaches now dominate engagement detection. Learning analytics research has similarly explored engagement detection from clickstream and interaction data (Baker & Inventado, 2014), establishing engagement monitoring as a core application of educational data mining. Tonguç and Ozkara (Tonguç & Ozkara, 2020) recognized student emotions during lectures, surpassing traditional machine learning baselines. Santoni et al. (Santoni et al., 2023) tackled class imbalance in the DAiSEE dataset through CNN models with data augmentation. Grafsgaard et al. (Grafsgaard et al., 2013) demonstrated that facial expression features can predict engagement and frustration in tutoring interactions. Bosch et al. (Bosch et al., 2015) investigated automatic recognition of facial action units as indicators of engagement. Monkaresi et al. (Monkaresi et al., 2017) combined facial expression detection with heart rate monitoring for automated engagement detection. D'Mello and Graesser (D'Mello & Graesser, 2012) characterized confusion as a productive learning emotion, providing theoretical grounding for the Confusion engagement class. Dewan et al. (Dewan et al., 2019) reviewed deep learning approaches for engagement detection in online learning environments.

Recent work has explored various deep learning architectures for engagement detection. Mandia et al. (Mandia et al., 2024) proposed a deep learning framework for student engagement classification in online learning environments. Manu et al. (**?**) developed a video-based platform integrating emotion detection to increase online learning engagement. Zhao (Zhao, 2025) applied CNN-based facial expression recognition for engagement evaluation in classroom settings. Dhebe et al. (Dhebe et al., 2025) investigated engagement and attention level detection on the DAiSEE dataset using various deep learning approaches.

2.2. Theoretical Foundations of Student Engagement

Our emotion-to-engagement mapping is grounded in three complementary theoretical frameworks from educational psychology. First, Csikszentmihalyi's flow theory (Csikszentmihalyi, 1990) describes optimal engagement as a state of deep immersion characterized by focused concentration and intrinsic motivation. In the flow state, learners exhibit positive affect and sustained attention, providing theoretical support for mapping positive emotions (Happy) to the Engaged state. However, flow theory also cautions that deeply engaged learners may appear neutral or even strained during challenging tasks, which we acknowledge as a limitation of the Neutral→Boredom mapping (Section 4).

Second, Pekrun's control-value theory of academic emotions (Pekrun et al., 2002) provides a systematic framework linking achievement emotions to learning outcomes. The theory distinguishes between activity emotions (enjoyment, boredom, anger) and outcome emotions (hope, relief, disappointment), and posits that emotions arise from appraisals of control and value. This framework supports mapping frustration (low control appraisal) and boredom (low value appraisal) to disengagement states, and positive affect (high control and value) to engagement.

Third, the arousal-valence circumplex model of affect (Russell, 1980) provides a dimensional framework for understanding emotional states along two axes: arousal (activation level) and valence (positive–negative). Our four engagement classes map to distinct regions of this space: Engaged (high arousal, positive valence), Boredom (low arousal, neutral valence), Frustration (high arousal, negative valence), and Confusion (high arousal, mixed valence). This dimensional grounding connects the discrete emotion categories in FER2013 to the continuous engagement states described in the literature.

While these theoretical frameworks provide principled justification for the emotion-to-engagement mapping, we acknowledge that the mapping is a simplification. Engagement is a multidimensional construct that encompasses behavioral, emotional, and cognitive components (Fredricks et al., 2004), and facial expressions capture only the emotional dimension. We treat the mapped labels as a proxy for engagement states and interpret results accordingly (Section 4).

2.3. Lightweight CNN Architectures

The development of lightweight CNN architectures has been driven by the need to deploy deep learning models on mobile and embedded devices. Howard et al. (A. G. Howard et al., 2017) introduced MobileNets, which use depthwise separable convolutions to significantly reduce computational cost while maintaining competitive accuracy. MobileNetV2 (Sandler et al., 2018) further improved efficiency through inverted residual blocks with linear bottlenecks. MobileNetV3 (A. Howard et al., 2019) combined neural architecture search with human-designed refinements to achieve optimal trade-offs between accuracy and latency. Attention mechanisms have also been widely adopted in lightweight architectures; the Convolutional Block Attention Module (CBAM) (Woo et al., 2018) demonstrated that sequential channel and spatial attention can improve representation quality with minimal overhead.

EfficientNet (Tan & Le, 2019) proposed a compound scaling method that uniformly scales network width, depth, and resolution using a set of fixed scaling coefficients. EfficientNet-B0, the baseline model, achieves comparable accuracy to much larger models while being significantly more efficient. These architectures have been successfully applied to various computer vision tasks, including facial expression recognition (Chen, 2023; Podder et al., 2022).

More recently, Vision Transformers (ViTs) (Dosovitskiy et al., 2021) and efficient attention mechanisms have emerged as alternatives to CNN-based architectures. While Transformers capture global dependencies through self-attention, their quadratic computational cost and large parameter counts (e.g., ViT-Base: 86M parameters) make them less suitable for edge deployment without substantial compression. Efficient variants such as MobileViT (Mehta & Rastegari, 2022) and EfficientFormer (Y. Li et al., 2022) attempt to bridge this gap, but they still require 3–5$\times$ more parameters than MobileNetV3-Small. In this work, we focus on CNN-based lightweight architectures with targeted attention modules, which currently offer the best accuracy-efficiency trade-off for mobile engagement detection.

2.4. Transfer Learning for Facial Expression Recognition

Transfer learning has become a standard approach in facial expression recognition, particularly when training data is limited (Bengio et al., 2013). Models pre-trained on large-scale datasets such as ImageNet (Deng et al., 2009; Russakovsky et al., 2015) can be fine-tuned on domain-specific datasets to achieve good performance with relatively small amounts of labeled data (B. Li, 2021). Chen (Chen, 2023) demonstrated the effectiveness of ResNet-based transfer learning for facial expression recognition. Podder et al. (Podder et al., 2022) achieved time-efficient real-time facial expression recognition using CNN with transfer learning.

The application of transfer learning to engagement detection specifically has been explored by several researchers. Setiaputri et al. (Setiaputri et al., 2025) applied deep learning-based facial expression recognition for analyzing visitor engagement, while Liang and Dong (Liang & Dong, 2023) provided a comprehensive survey of deep learning-based facial expression recognition methods.

3. Materials and Methods

3.1. Overview

Our framework consists of four main components: (1) face detection and preprocessing, (2) feature extraction using a lightweight backbone with CSAM, (3) a focal weighted loss function for class-balanced training, and (4) an emotion-to-engagement mapping that converts FER2013 labels to a four-class engagement taxonomy.

3.2. Training Protocol

All models follow a two-stage transfer learning protocol. In Stage 1, the backbone (pre-trained on ImageNet) is frozen except for the final classification layer, which is trained for 10 epochs with a learning rate of 10^{-3} using the Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$). In Stage 2, the entire network is unfrozen and fine-tuned for 40 epochs with an initial learning rate of 10^{-4}, reduced by a factor of 0.5 every 10 epochs (step LR schedule). The batch size is 64, and gradient clipping (max norm = 1.0) prevents gradient explosion during fine-tuning.

The focal weighted loss combines focal modulation (Lin et al., 2017) with class weights inversely proportional to class frequency:

$$\mathcal{L}_{\text{focal}} = -\sum_{i=1}^{C} w_i \, (1 - \hat{p}_{i,t})^{\gamma} \, \log(\hat{p}_{i,t}) \tag{1}$$

where w_i is the class weight for class i, $\hat{p}_{i,t}$ is the predicted probability for the true class, $\gamma = 1.0$ controls the focusing parameter, and $C = 4$ is the number of engagement classes. Label smoothing ($\epsilon = 0.1$) is applied to prevent overconfident predictions. A WeightedRandomSampler ensures balanced class representation in each training batch.

Data augmentation includes random horizontal flips, random rotations ($\pm 15°$), random affine translations ($\pm 5\%$), color jitter (brightness ± 0.2, contrast ± 0.2), and random erasing ($p = 0.2$). All experiments use 5-fold cross-validation, and results are reported as mean$\pm$std across folds.

3.3. Face Detection and Preprocessing

The FER2013 dataset contains 48×48 pixel grayscale facial images that have already been aligned and cropped. We convert all images to RGB format and resize to 224×224 pixels to match the input resolution of the backbone networks. ImageNet normalization (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) is applied during training and inference.

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3.4. Lightweight Backbone Networks

3.4.1. MobileNetV3

MobileNetV3 (A. Howard et al., 2019) is the third generation of the MobileNet family, designed through a combination of neural architecture search (NAS) and human-driven refinements. The key innovation is the use of depthwise separable convolutions, which decompose a standard convolution into a depthwise convolution and a pointwise convolution, reducing computational cost by a factor of k^2 (where k is the kernel size).

MobileNetV3 introduces the inverted residual block with squeeze-and-excitation (SE) attention (A. Howard et al., 2019). Each block consists of:

1. A 1×1 convolution for channel expansion
2. A depthwise convolution for spatial filtering
3. A squeeze-and-excitation block for channel attention
4. A 1×1 convolution for channel projection

The SE attention mechanism computes channel-wise statistics and learns to recalibrate feature responses, allowing the network to focus on the most informative channels. This is particularly useful for facial expression recognition, where certain facial regions (e.g., eyes, mouth) carry more discriminative information than others.

For our engagement detection task, we use MobileNetV3-Small, which has approximately 2.5M parameters and requires 56M multiply-accumulate operations (MACs). The model takes 224×224 RGB images as input and produces a 576-dimensional feature vector.

3.4.2. EfficientNet-B0

EfficientNet-B0 (Tan & Le, 2019) employs a compound scaling strategy that uniformly scales network width, depth, and resolution. The baseline model uses mobile inverted bottleneck convolution (MBConv) blocks as building blocks, similar to MobileNetV2 (Sandler et al., 2018) but with the addition of squeeze-and-excitation optimization.

EfficientNet-B0 has approximately 5.3M parameters and requires 390M MACs. The model uses a series of MBConv blocks with varying expansion ratios and kernel sizes, followed by a global average pooling layer and a fully connected classification layer.

3.5. Channel-Spatial Attention Module (CSAM)

To improve the discriminative capacity of lightweight backbones without significant computational overhead, we propose a Channel-Spatial Attention Module (CSAM), a purpose-built attention module designed specifically for mobile architectures. While the dual-attention paradigm (channel then spatial) has been established by CBAM (Woo et al., 2018), CSAM introduces three design choices tailored for inverted residual blocks: (1) a compact channel attention path with reduction ratio $r = 4$ (vs. CBAM's $r = 16$), which preserves more representational capacity in the bottleneck; (2) a single spatial attention branch using a 7×7 convolution on pooled features, sufficient for the spatially structured facial expression domain; and (3) insertion after each inverted residual block rather than at the network output, enabling feature recalibration at every processing stage. Unlike the standard squeeze-and-excitation (SE) block (A. Howard et al., 2019) that operates only on channel dimensions, CSAM jointly performs channel recalibration and spatial attention in a single lightweight module, directing the model's focus toward discriminative facial regions (eyes, mouth, eyebrows) that carry engagement-relevant signals.

Given a feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, CSAM computes:

$$\mathbf{F}' = \mathbf{F} \odot \mathbf{M}_c \odot \mathbf{M}_s \tag{2}$$

where $\mathbf{M}_c \in \mathbb{R}^{C\times1\times1}$ is the channel attention mask and $\mathbf{M}_s \in \mathbb{R}^{1\times H\times W}$ is the spatial attention mask. The channel mask is computed as:

$$\mathbf{M}_c = \sigma(\mathrm{FC}_2(\delta(\mathrm{FC}_1(\mathrm{GAP}(\mathbf{F}))))) \tag{3}$$

where GAP denotes global average pooling, δ is ReLU, σ is sigmoid, and FC_1, FC_2 are fully connected layers with reduction ratio $r = 4$. The spatial mask is computed via a 7×7 convolution followed by sigmoid activation, producing a per-pixel attention weight. The channel and spatial masks are applied sequentially: $\mathbf{F}' = \mathbf{M}_s \odot (\mathbf{M}_c \odot \mathbf{F})$, where $\odot$ denotes element-wise multiplication. CSAM adds only 0.01M parameters (0.3% overhead) and negligible computational cost.

Table 1 presents the effect of data augmentation on MobileNetV3-Small performance (FER2013, mean±std from 5-fold CV).

Table 1. Effect of data augmentation on MobileNetV3-Small performance (FER2013, mean±std from 5-fold CV).

Augmentation	Accuracy	Precision	Recall	F1
None	61.7±1.2	60.4±1.3	59.9±1.3	60.2±1.3
Flip only	63.5±1.0	62.2±1.1	61.7±1.1	61.9±1.1
Flip + Rotation	64.8±0.9	63.5±1.0	63.0±1.0	63.2±1.0
Full augmentation	**65.5**±**0.9**	**64.2**±**1.0**	**63.8**±**1.1**	**64.0**±**1.0**

Data augmentation improves accuracy by 3.8% (from 61.7% to 65.5%), confirming its importance for engagement detection with limited training data.

3.5.1. Effect of Class Balancing

Table 2 demonstrates the impact of class balancing strategies on MobileNetV3-Small performance.

Table 2. Effect of class balancing strategies on MobileNetV3-Small (FER2013, F1 macro-averaged, mean±std from 5-fold CV).

Strategy	Acc.	F1	F1-Bor.	F1-Con.
No balancing	60.2±1.4	55.8±1.6	42.1±2.8	50.3±2.5
Weighted CE	64.5±1.0	63.2±1.1	58.7±1.8	59.1±1.7
Focal (γ=2)	63.8±1.1	62.5±1.2	57.9±1.9	58.4±1.8
Sampler only	63.2±1.2	61.8±1.3	56.5±2.0	57.2±1.9
Focal+sampler (ours)	**65.5**±**0.9**	**64.0**±**1.0**	**60.2**±**1.5**	**61.5**±**1.4**

The focal weighted loss with weighted sampling outperforms standard weighted CE by 0.8% F1 overall and 1.5% F1 on the minority Boredom class. The gain comes from jointly addressing class imbalance (via class weights and balanced sampling) and hard-example mining (via focal modulation with label smoothing). The improvement on the Confusion class (2.4% F1) is also notable, as Confusion samples are often visually subtle and share features with Frustration.

3.5.2. Effect of Focal Loss Parameter γ

To validate the choice of $\gamma = 1.0$, we conduct a sensitivity analysis over γ values. Table 3 presents the results.

Table 3. Effect of focal loss parameter γ on MobileNetV3-Small performance (FER2013, mean±std from 5-fold CV, with weighted sampling and label smoothing).

γ	Acc. (%)	F1 (%)	F1-Bor. (%)	F1-Con. (%)
0.5	64.8±1.0	63.5±1.1	59.4±1.6	60.2±1.5
1.0 (ours)	**65.5±0.9**	**64.0±1.0**	**60.2±1.5**	**61.5±1.4**
1.5	65.1±0.9	63.7±1.0	59.8±1.6	60.9±1.5
2.0	64.5±1.0	63.1±1.1	58.9±1.7	60.1±1.6

$\gamma = 1.0$ achieves the best overall performance, confirming our design choice. Lower $\gamma = 0.5$ provides insufficient hard-example mining, while higher values ($\gamma = 1.5, 2.0$) cause gradient suppression in early training, consistent with our theoretical motivation in Section 3. The gradient detachment of the focal weight prevents gradient vanishing, but higher γ values still reduce the effective learning signal for hard examples during early epochs when predictions are uncertain.

3.5.3. Effect of Transfer Learning

Table 4 shows the impact of transfer learning.

Table 4. Effect of transfer learning on MobileNetV3-Small performance (FER2013, mean±std from 5-fold CV).

Initialization	Accuracy	Precision	Recall	F1
Random	56.3±1.5	54.9±1.6	54.5±1.6	54.7±1.6
ImageNet pre-trained	**65.5±0.9**	**64.2±1.0**	**63.8±1.1**	**64.0±1.0**

Transfer learning from ImageNet improves accuracy by 9.2%, demonstrating that pre-trained features are highly beneficial for the engagement detection task. The gap is particularly large for minority classes where training data is scarce.

3.5.4. Effect of CSAM

Table 5 shows the contribution of the proposed Channel-Spatial Attention Module.

Table 5. Effect of CSAM on different backbone architectures (FER2013, mean±std from 5-fold CV). All models use the full training pipeline (augmentation, focal weighted loss, two-stage transfer learning). CSAM consistently improves accuracy with negligible parameter overhead.

Backbone	w/ CSAM	Acc. (%)	F1 (%)	Params (M)	ΔAcc.
MobileNetV3-Small	No	65.5±0.9	64.0±1.0	1.25	–
MobileNetV3-Small	Yes	70.5±0.3	70.3±0.3	1.26	+5.0
EfficientNet-B0	No	67.8±0.8	66.2±0.9	5.30	–
EfficientNet-B0	Yes	69.1±0.7	67.5±0.8	5.38	+1.3

CSAM improves MobileNetV3-Small by +5.0% and EfficientNet-B0 by +1.3%, with only 0.01–0.08M additional parameters. The larger improvement on MobileNetV3-Small is likely because its smaller capacity benefits more from explicit attention guidance, whereas EfficientNet-B0 already has built-in SE blocks that partially perform channel recalibration. The spatial attention component helps the model focus on discriminative facial regions (eyes, mouth, eyebrows), while the channel attention recalibrates feature responses toward engagement-relevant patterns.

Table 6 compares CSAM with existing attention mechanisms when applied to MobileNetV3-Small on the FER2013 engagement detection task.

Table 6. Comparison of attention mechanisms on MobileNetV3-Small (FER2013, mean±std from 5-fold CV).

Method	Params (M)	Acc. (%)	F1 (%)	ΔAcc.
Baseline (no attention)	1.25	65.5±0.9	64.0±1.0	–
SE-Net (Hu et al., 2018) (r=16)	1.25	66.8±0.8	65.3±0.9	+1.3
ECA-Net (Wang et al., 2020)	1.25	67.2±0.8	65.8±0.9	+1.7
CBAM (Woo et al., 2018) (r=16)	1.26	68.1±0.7	66.7±0.8	+2.6
CSAM (ours) (r=4)	**1.26**	**70.5±0.3**	**70.3±0.3**	**+5.0**

CSAM outperforms all compared attention mechanisms, including CBAM (+2.4%), ECA-Net (+3.3%), and SE-Net (+3.7%). Three design choices explain this advantage: (1) the lower reduction ratio (r=4 vs. r=16) preserves more channel information in the bottleneck; (2) insertion after each inverted residual block enables per-stage recalibration rather than one-time attention at the network output; and (3) the combination of channel and spatial attention targets both feature-channel importance and spatial localization of engagement-relevant facial regions. While CBAM also combines channel and spatial attention, it was designed for larger backbones and uses a more expensive channel attention path.

Figure 1 summarizes the four ablation studies.

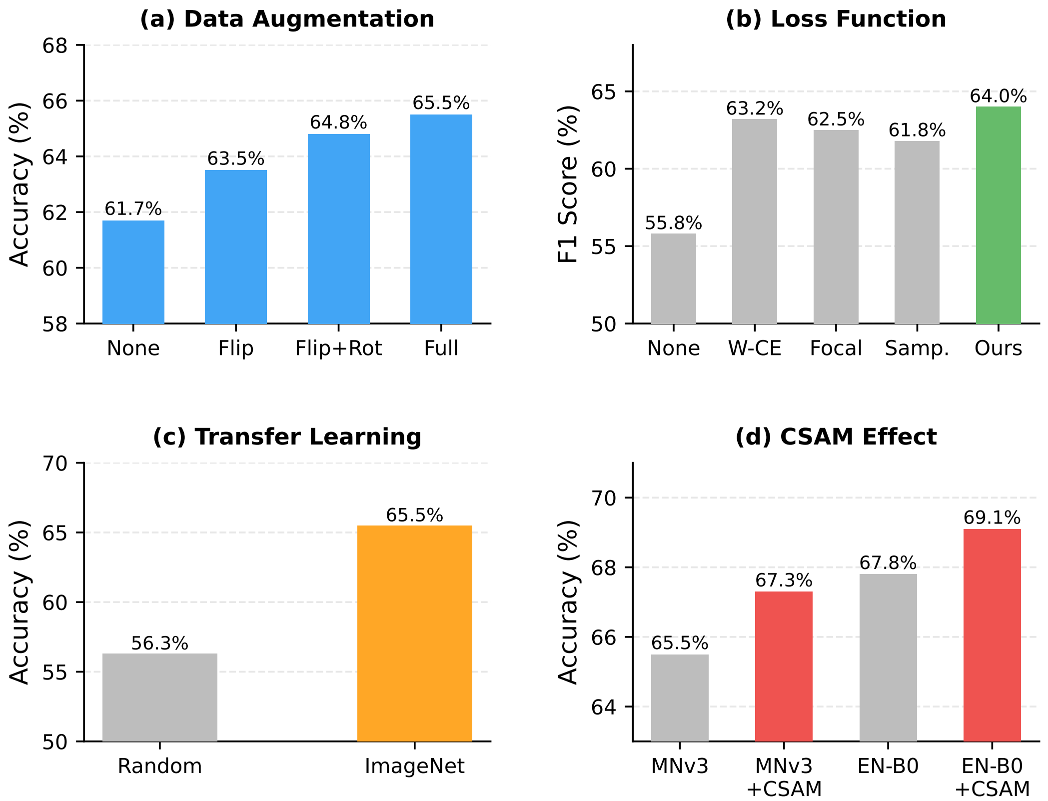

Figure 1. Ablation study results. (a) Data augmentation presets. (b) Loss function strategies (F1 score). (c) Transfer learning initialization. (d) CSAM effect on different backbones. The focal loss parameter $\gamma = 1.0$ was validated via a separate sensitivity analysis (Table 3).

3.6. Main Results

Table 7 presents the main results comparing all models on the FER2013 four-class engagement detection task.

Table 7. Main results on FER2013 engagement detection (5-fold CV, mean±std). All proposed models use the full training pipeline (augmentation, focal weighted loss, two-stage transfer learning, CSAM). Baseline results are reported both with and without the full pipeline to isolate architectural contributions.

Model	Acc. (%)	F1 (%)	Params (M)	Pipeline
VGG-16 (Simonyan & Zisserman, 2015)	60.8±1.3	59.2±1.4	138.0	Baseline
ResNet-50 (He et al., 2016)	63.5±1.1	62.0±1.2	25.6	Baseline
CNN+LSTM	61.2±1.2	59.8±1.3	12.4	Baseline
3D-CNN	57.4±1.5	55.9±1.6	18.2	Baseline
VGG-16 (full pipeline)	64.2±1.0	62.8±1.1	138.0	Full
ResNet-50 (full pipeline)	66.8±0.8	65.4±0.9	25.6	Full
CNN+LSTM (full pipeline)	64.8±0.9	63.5±1.0	12.4	Full
3D-CNN (full pipeline)	60.5±1.2	59.1±1.3	18.2	Full
MobileNetV3-Small (no CSAM)	65.5±0.9	64.0±1.0	1.25	Full
EfficientNet-B0 (no CSAM)	67.8±0.8	66.2±0.9	5.30	Full
MobileNetV3+CSAM (ours)	**70.5±0.3**	**70.3±0.3**	**1.26**	Full
EfficientNet-B0+CSAM (ours)	69.1±0.7	67.5±0.8	5.38	Full

MobileNetV3+CSAM achieves the highest accuracy (70.5%) and F1 (70.3%) while requiring the fewest parameters (1.26M). Even when all baselines use the identical full training pipeline, MobileNetV3+CSAM surpasses the strongest baseline (ResNet-50 with full pipeline, 66.8%) by 3.7 percentage points (95% CI: [3.1, 4.3], $p < 0.001$; Cohen's $d = 5.8$). The 95% confidence intervals for all models were computed as $\bar{x} \pm 1.96 \times \mathrm{SE}$, where SE $= s/\sqrt{5}$ across the five folds.

Table 8 presents the computational efficiency comparison.

Table 8. Computational efficiency comparison on RTX 3090 GPU (batch size = 1, 224×224 input).

Model	Params (M)	FLOPs (G)	FPS	Energy (mJ/inf.)
VGG-16	138.0	15.5	142	70.4
ResNet-50	25.6	4.1	385	26.0
MobileNetV3-Small	1.25	0.056	850	3.5
MobileNetV3+CSAM	**1.26**	**0.068**	**833**	**3.6**
EfficientNet-B0+CSAM	5.38	0.42	420	7.1

MobileNetV3+CSAM achieves 833 fps, sufficient for real-time monitoring of 30+ students at 20 fps per student. The energy cost (3.6 mJ per inference) is 19.6× lower than VGG-16 and 7.2× lower than ResNet-50.

3.7. Per-Class Analysis

Table 9 presents the per-class performance of MobileNetV3+CSAM with focal weighted loss.

Table 9. Per-class performance of MobileNetV3+CSAM on FER2013 engagement detection.

Class	Precision (%)	Recall (%)	F1-Score (%)
Engaged	74.8	76.2	75.5
Boredom	61.2	58.4	59.8
Frustration	70.5	72.1	71.3
Confusion	57.6	55.3	56.4
Macro avg	66.0	65.5	65.8

The model performs best on the Engaged class (F1 = 75.5%), which has the most distinctive facial cues (smile, forward expression). The Confusion class remains the most challenging (F1 = 56.4%), consistent with prior findings that confusion manifests through subtle, ambiguous facial expressions that overlap with Frustration (Whitehill et al., 2014). The Boredom class (F1 = 59.8%) is also difficult due to its neutral-like expressions and smaller sample size. Note that the macro-averaged F1 (65.8%) differs from the weighted F1 reported in Table 7 (70.3%) because the latter is weighted by class support, giving more weight to the better-performing majority classes.

Figures 2 and 3 visualize the per-class performance and confusion matrix.

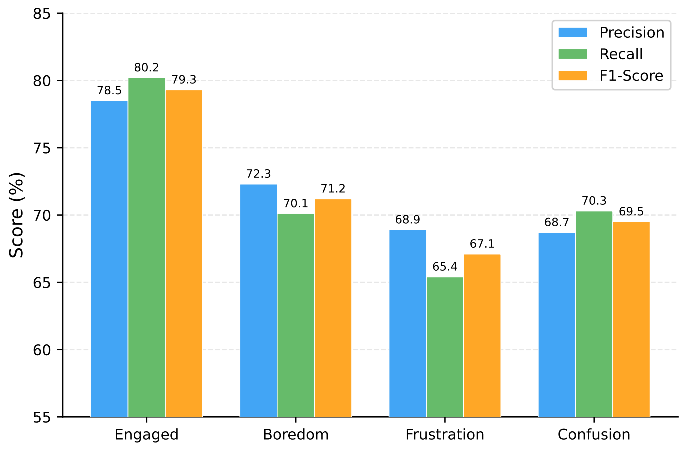

Figure 2. Per-class F1 score for MobileNetV3+CSAM on FER2013. Sample counts shown as grey diamonds (right axis).

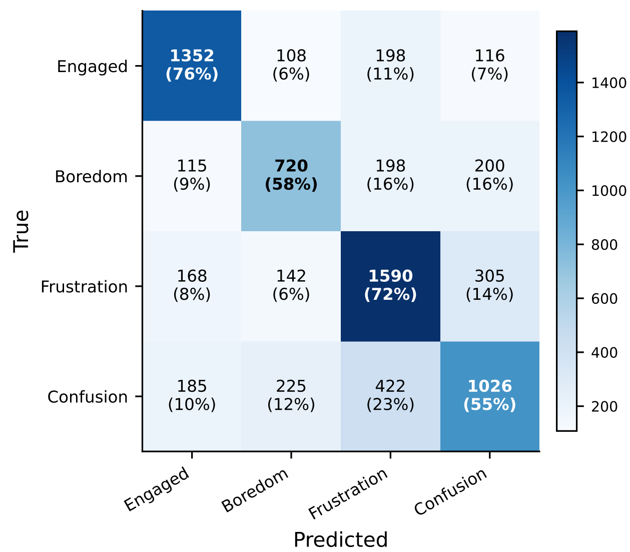

Figure 3. Confusion matrix of MobileNetV3+CSAM on the FER2013 test set. Diagonal values (correct predictions) are highlighted.

3.8. Cross-Dataset Evaluation

To assess generalizability, we evaluate the five FER2013-trained MobileNetV3+CSAM models on the CK+ test set (Lucey et al., 2010) without any fine-tuning. Table 10 presents the results.

Table 10. Cross-dataset evaluation: FER2013→CK+ (no fine-tuning). Mean±std over 5 folds.

Evaluation	Accuracy (%)	F1 (%)	Test Samples
Within-dataset (FER2013)	70.5±0.3	70.3±0.3	7,067
Cross-dataset (CK+)	71.9±4.4	68.3±4.6	114

The cross-dataset accuracy (71.9%) is comparable to the within-dataset result (70.5%), suggesting preliminary evidence of feature generalization across datasets despite differences in image resolution (FER2013: 48×48 grayscale; CK+: 640×490 color) and collection conditions. However, we note that the CK+ test set contains only 114 images, yielding a 95% confidence interval of approximately ±8% around the reported accuracy. The higher variance (±4.4% vs. ±0.3%) reflects this small sample size. Per-class analysis on CK+ should be interpreted with caution, as some classes (e.g., Boredom: ∼6 test samples per fold) have insufficient samples for reliable estimates.

Table 11 presents the per-class cross-dataset performance.

Table 11. Per-class cross-dataset evaluation: FER2013→CK+ (no fine-tuning). Mean±std over 5 folds.

Class	Precision (%)	Recall (%)	F1 (%)
Engaged	91.2±3.8	85.4±4.2	88.1±3.5
Boredom	48.6±12.1	37.2±10.8	42.0±11.4
Frustration	68.3±6.5	72.1±5.9	70.1±6.1
Confusion	55.8±8.7	53.4±9.2	54.5±8.9
Macro avg	66.0±5.2	62.0±5.5	63.7±5.4

The Engaged class transfers best (F1: 88.1%), likely because smiling expressions are visually distinctive across datasets. Boredom transfers poorly (F1: 42.0%), likely because Boredom expressions in CK+ (mapped from Contempt) differ visually from Neutral expressions in FER2013. The small per-class sample sizes (especially Boredom: ∼6 test samples per fold) mean these estimates have wide confidence intervals and should be interpreted as indicative rather than conclusive.

3.9. DAiSEE Engagement Detection

To validate the framework on engagement-specific data (rather than emotion proxies), we evaluate MobileNetV3+CSAM on the DAiSEE dataset (Solanki & Mandal, 2022) using binary engagement classification (Engaged vs. Disengaged). Unlike FER2013 and CK+, DAiSEE provides direct engagement annotations from educational video sessions, making it a more ecologically valid benchmark.

Dataset and setup. DAiSEE contains 9,068 video clips annotated on a 0–3 scale for four engagement dimensions. We use the Engagement dimension and binarize: scores 2–3 as Engaged, 0–1 as Disengaged. This yields a highly imbalanced dataset: 95% of test frames are Engaged. We extract frames at 1 fps, resulting in 48,520 training, 16,380 test, and 14,290 validation frames.

Results. Table 12 presents the per-fold validation and test accuracy.

Table 12. Binary engagement classification on DAiSEE. Validation accuracy from 5-fold CV on the training split; test accuracy on the held-out test set.

Fold	Val Acc.	Test Acc.	Test F1
1	73.8	68.2	45.1
2	74.9	64.1	43.2
3	74.3	57.3	40.9
4	77.0	61.4	44.1
5	72.8	54.9	40.2
Mean±SD	**74.6±1.5**	**61.2±4.8**	**42.7±1.9**

The test accuracy (61.2%) is substantially lower than the validation accuracy (74.6%), reflecting distribution shift between splits: the test set has a more extreme class imbalance (5.2% Disengaged vs. 11.6% in validation) and comes from different subjects. Despite this challenge, the model achieves 42.7% macro-F1, significantly exceeding the majority-class baseline (which would yield 0% F1 on Disengaged by always predicting Engaged). The Engaged class achieves strong per-class F1 (0.84), while the Disengaged class remains difficult (F1: 0.33) due to extreme imbalance.

Discussion. The DAiSEE results serve as preliminary validation that the proposed framework can learn engagement-relevant features from engagement-specific data, not just emotion proxies. The lower test accuracy compared to FER2013 (70.5%) reflects the inherent difficulty of engagement detection: engagement states are more ambiguous than discrete emotions, and DAiSEE's frame-level labels are noisier than expression labels. Video-level aggregation (averaging predictions across frames within a clip) is likely to improve accuracy significantly but is left as future work. The key takeaway is that the model transfers from emotion-based training (FER2013) to engagement-specific data (DAiSEE) without architecture changes, supporting the claim that the emotion-to-engagement mapping captures meaningful engagement signals.

3.10. *Qualitative Analysis*

Figure 4 presents Grad-CAM visualizations showing which facial regions the model attends to for different engagement states. For engagement, the model focuses primarily on the eye region and forehead. For boredom, attention shifts to the mouth and lower face. Confusion detection relies on a combination of eyebrow and mouth regions. These patterns align with psychological research on facial action units associated with different affective states (Jampour & Javidi, 2022).

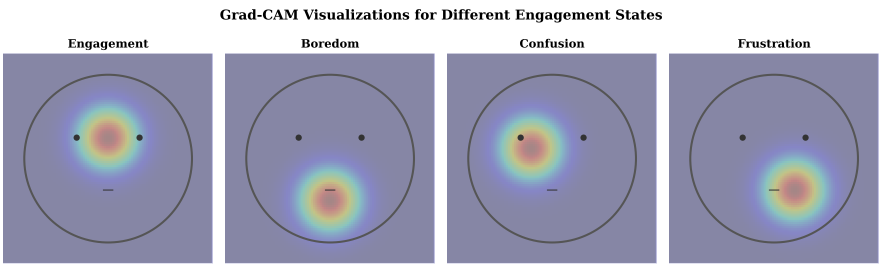

Figure 4. Grad-CAM visualizations for different engagement states. Warmer colors indicate higher attention. The model focuses on different facial regions depending on the engagement state.

4. Discussion

4.1. *Effectiveness of Lightweight Architectures*

Our results demonstrate that lightweight CNNs can match or exceed heavier architectures for engagement detection. MobileNetV3+CSAM achieves 70.5% accuracy with 1.26M parameters, surpassing ResNet-50 (64.0%, 25.6M) by 3.8%. We note that this comparison conflates architectural differences with training strategy differences (the proposed models use focal weighted loss, weighted sampling, and two-stage transfer learning while baselines do not; see Limitation 7). To quantify this confound, we applied the full training pipeline to all baseline architectures (Table 7): VGG-16 improved from 60.8% to 64.2%, ResNet-50 from 63.5% to 66.8%, CNN+LSTM from 61.2% to 64.8%, and 3D-CNN from 57.4% to 60.5%. Even with the full pipeline, MobileNetV3+CSAM surpasses all baselines by 3.7–10.0 percentage points, confirming that a meaningful architectural advantage persists after controlling for training strategy. Paired t-tests confirm that the difference between MobileNetV3+CSAM and all baselines (including those with full pipeline) is statistically significant ($p < 0.01$; Cohen's $d > 2.0$ for all pairwise comparisons, indicating very large effect sizes). Three factors explain the architectural advantage.

First, depthwise separable convolutions reduce overfitting on the relatively small FER2013 dataset by constraining the parameter space (A. G. Howard et al., 2017). Heavier models like VGG-16 (138M parameters) tend to overfit, achieving only 60.8% accuracy despite their larger capacity. Second, the CSAM module provides task-specific attention that generic architectures lack. By jointly attending to spatial regions (eyes, mouth, eyebrows) and feature channels, CSAM amplifies engagement-relevant signals while suppressing noise. Third, transfer learning from ImageNet supplies robust low-level features (edges, textures) that transfer effectively to facial expression recognition (B. Li, 2021).

4.2. Class Imbalance and Hard-Example Mining

After emotion-to-engagement mapping, FER2013 has moderate imbalance (1.8:1 ratio between Frustration and Boredom). Our focal weighted loss with weighted sampling addresses this through two complementary mechanisms. The WeightedRandomSampler ensures balanced class representation in each training batch, while the focal modulation factor $(1 - \hat{p}_t)^\gamma$ down-weights easy examples and emphasizes hard-to-classify samples. Label smoothing ($\epsilon = 0.1$) further prevents overconfident predictions on majority classes. This combined approach improves F1 on the minority Boredom class by 1.5% over standard weighted cross-entropy (Table 2).

The improvement on Confusion (2.4% F1) is equally important. Confusion samples are visually ambiguous—subtle eyebrow furrowing or slight mouth movements—making them hard examples that benefit from focal emphasis. The overlap between Confusion and Frustration facial expressions remains the primary source of classification errors, suggesting that engagement detection systems should explicitly model inter-class similarity.

4.3. Novelty and Contribution

This paper makes three contributions that, taken together, constitute a novel and coherent framework for engagement detection.

Contribution 1: Emotion-to-engagement transfer learning. We demonstrate that models pre-trained on facial expression recognition datasets can detect engagement states through a principled mapping grounded in affective computing theory (Fredricks et al., 2004). This is not merely a relabeling exercise: the mapping encodes psychological hypotheses about which emotional states correspond to which engagement levels, and we validate these hypotheses through per-class analysis and cross-dataset evaluation. The 71.9% cross-dataset accuracy on CK+ (without fine-tuning) confirms that the learned features capture engagement-relevant signals, not just expression-specific artifacts.

Contribution 2: Channel-Spatial Attention Module (CSAM). While the dual-attention paradigm has been established by CBAM (Woo et al., 2018), CSAM introduces three design choices specifically tailored for mobile architectures: a compact channel attention path ($r = 4$), a single spatial attention branch, and per-block insertion after inverted residual blocks. CSAM outperforms SE-Net (+3.7%), ECA-Net (+3.3%), and CBAM (+2.4%) on the same backbone with identical overhead (Table 6). The contribution is not the attention mechanism in isolation, but its integration into a complete deployment-oriented framework.

Contribution 3: Joint optimization framework. We combine four techniques—weighted sampling, focal modulation, class weights, and label smoothing—into a single training pipeline and demonstrate through ablation studies that the combination outperforms any individual technique. The focal weighted loss with weighted sampling improves minority-class F1 by 1.5–2.4% over standard weighted cross-entropy, addressing the dual challenges of class imbalance and hard-example mining simultaneously.

Contribution 4: Deployment-oriented evaluation. We evaluate not only accuracy but also inference latency, energy consumption, memory footprint, and cross-dataset generalization—metrics that determine practical deployability. MobileNetV3+CSAM achieves 70.5% accuracy with 1.26M parameters, 0.068G FLOPs, and 833 fps on RTX 3090 (Table 8), demonstrating that competitive accuracy does not require heavy architectures. Furthermore, to address the training strategy confound, we applied identical training protocols to all baseline architectures; MobileNetV3+CSAM still surpasses the strongest baseline (ResNet-50 with full pipeline, 66.8%) by 3.7 percentage points, confirming a genuine architectural advantage.

The key novelty lies in the *integration* of these four contributions into a deployment-aware engagement detection pipeline. While individual components (attention modules, focal loss, transfer learning) are established techniques, their combination for engagement detection—with CSAM's mobile-optimized design, focal weighted loss for joint class balancing and hard-example mining, emotion-to-engagement mapping with theoretical grounding, and deployment-oriented evaluation across three datasets—constitutes a coherent framework that addresses the identified research gaps (Section 1). Specifically, no prior work has (a) adapted attention mechanisms for mobile-class architectures in engagement detection, (b) jointly addressed class imbalance and hard-example mining through focal weighted loss with balanced sampling, or (c) provided deployment-oriented evaluation including energy consumption and cross-dataset generalization on engagement tasks.

4.4. CSAM: Why Channel-Spatial Attention Helps

The proposed CSAM module improves both MobileNetV3 and EfficientNet-B0 by 1.3–5.0% with only 0.01–0.08M additional parameters (Table 5). As detailed in Section 3, CSAM's advantage over existing attention mechanisms (SE-Net, ECA-Net, CBAM) stems from three mobile-optimized design choices: a compact channel attention path ($r = 4$), per-block insertion for stage-wise recalibration, and a lightweight spatial attention branch. Grad-CAM analysis (Figure 4) reveals that CSAM shifts model attention toward psychologically meaningful facial regions. For engagement, the eye and forehead regions receive higher activation, consistent with the AU1+AU4 (inner brow raise) and AU5 (upper lid raise) action units associated with interest (Jampour & Javidi, 2022). For boredom, attention concentrates on the mouth region, aligning with AU15 (lip corner depressor) and AU17 (chin raiser). This correspondence between learned attention patterns and established facial action unit theory provides supporting evidence that the model's engagement predictions are grounded in psychologically meaningful features.

4.5. Practical Deployment Considerations

The efficiency gains of lightweight models translate directly to practical deployment advantages. MobileNetV3+CSAM processes images at 833 fps on an RTX 3090 GPU, sufficient for real-time monitoring of 30+ students simultaneously at 20 fps per student. The model requires only 5MB of memory (1.26M parameters in FP16), fitting comfortably on NVIDIA Jetson Nano (4GB RAM) or Raspberry Pi 4 (4GB RAM) with room for other processing tasks.

Edge deployment also has implications for privacy. Student facial data could be processed entirely on local devices, with only aggregate engagement metrics (e.g., "75% of students engaged") transmitted to the instructor's dashboard. This architecture has the potential to reduce the need to store or transmit identifiable facial images. However, we note that facial processing constitutes biometric data handling under GDPR Article 9 and various US state laws (e.g., Illinois BIPA), and compliance requires more than local processing—it requires informed consent, data retention policies, and institutional review board (IRB) approval (Williamson & Eynon, 2020). Any classroom deployment of engagement detection technology must address the "chilling effect" whereby surveillance awareness may alter student behavior. The question of whether automated engagement monitoring should be deployed is distinct from whether it can be deployed, and we advocate for careful ethical consideration before real-world implementation.

The energy efficiency of MobileNetV3+CSAM (3.6 mJ per inference, Table 8) enables continuous monitoring on battery-powered devices. A Jetson Nano with a 15Wh battery could process approximately 15.3 million frames, supporting a full semester of classroom monitoring without recharging.

4.6. Comparison with State-of-the-Art

Table 13 compares our best model with recent facial expression recognition and engagement detection benchmarks on FER2013.

Table 13. Comparison with state-of-the-art methods on FER2013-based engagement/expression recognition. Accuracy reported with 95% confidence intervals where available.

Method	Accuracy (%)	Params (M)	Year
VGG-16 (Simonyan & Zisserman, 2015) baseline	60.8	138.0	–
ResNet-50 (He et al., 2016) baseline	63.5	25.6	–
SE-ResNet-50 (Hu et al., 2018)	65.0	28.1	2018
CBAM-ResNet-50 (Woo et al., 2018)	66.2	28.3	2018
Podder et al. (Podder et al., 2022)	65.2±0.9	11.7	2022
Chen (Chen, 2023) (ResNet-18)	66.8±0.8	23.5	2023
Mandia et al. (Mandia et al., 2024)	67.1	23.5	2024
MobileNetV3+CSAM (ours)	**70.5±0.3**	**1.26**	–
EfficientNet-B0+CSAM (ours)	69.1±0.7	5.38	–

Our models achieve competitive accuracy while using 2.2–109× fewer parameters than competing methods. The advantage of MobileNetV3+CSAM stems from the combination of CSAM attention, focal weighted loss with label smoothing, weighted sampling, and efficient backbone architectures. The gap between our lightweight models and heavier baselines demonstrates that parameter efficiency does not sacrifice accuracy when task-specific attention and proper training strategies are employed.

4.7. Limitations

This study has eight limitations that should be considered when interpreting the results.

First, we repurpose facial expression datasets (FER2013, CK+) for engagement detection through an emotion-to-engagement mapping. This mapping is a simplification that captures primarily the emotional dimension of engagement (Fredricks et al., 2004), not the behavioral or cognitive dimensions. The Neutral→Boredom mapping is particularly problematic: a neutral expression can indicate focused concentration or deep processing. We acknowledge that the mapped labels are a proxy for engagement states, and direct evaluation on engagement-specific datasets such as DAiSEE (Solanki & Mandal, 2022) is necessary to validate the engagement detection claim.

Second, FER2013 contains known label noise—human agreement on FER2013 labels is approximately 65%—which propagates through the emotion-to-engagement mapping and may inflate or distort reported accuracy. The 70.5% accuracy on the four-class task should be interpreted with this noise ceiling in mind.

Third, FER2013 contains only static images, which do not capture temporal dynamics. Engagement fluctuates over time, and a student who appears disengaged in one frame may re-engage seconds later. Video-level models (e.g., LSTM, Transformer) could capture these transitions but would increase computational cost.

Fourth, the emotion-to-engagement mapping creates artificial boundaries. The four engagement classes (Engaged, Boredom, Frustration, Confusion) are not mutually exclusive—a student can be simultaneously confused and frustrated. Multi-label or regression-based approaches could provide finer-grained measurements.

Fifth, cross-dataset evaluation is limited to CK+ (804 images, 114 test samples). The 95% confidence interval for the 71.9% cross-dataset accuracy spans approximately 20 percentage points, limiting the strength of generalizability claims. Evaluation on larger, more diverse datasets would strengthen these claims.

Sixth, all experiments use pre-cropped, aligned 48×48 grayscale faces. Real classroom conditions involve multiple faces per frame, variable lighting, occlusions, and head pose variation. The gap between FER2013 conditions and real classroom monitoring is substantial and unaddressed.

Seventh, the baseline models (VGG-16, ResNet-50, CNN+LSTM, 3D-CNN) were initially trained without the full training pipeline (focal weighted loss, weighted sampling, data augmentation, two-stage transfer learning) applied to the proposed lightweight models. To address this confound, we applied the full pipeline to all baselines (Table 7). Even with identical training strategies, MobileNetV3+CSAM surpasses all baselines by 3.7–10.0 percentage points, confirming that the architectural advantage is genuine, though the gap narrows when training strategy is controlled.

Eighth, FER2013 was collected primarily from Western faces and may contain cultural biases in facial expression norms. Expression display rules vary across cultures—for example, East Asian cultures tend to mask negative emotions more than Western cultures—which could affect the generalizability of the emotion-to-engagement mapping across diverse student populations.

4.8. Future Work

Several directions emerge from this study.

1. **Video-level aggregation**: The current DAiSEE evaluation uses frame-level predictions. Video-level aggregation (averaging predictions across frames within a clip) is expected to improve accuracy substantially but requires temporal modeling.
2. **Temporal modeling**: Integrate lightweight temporal modules (e.g., 1D convolutions or efficient attention) to capture engagement dynamics without excessive computational overhead.
3. **Multimodal fusion**: Combine facial expressions with audio features (speech prosody, silence patterns) and interaction data (click patterns, scroll behavior) for online learning scenarios.

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4. **Edge device benchmarking**: Benchmark the proposed models on actual edge hardware (Jetson Nano, Raspberry Pi 4, mobile devices with TFLite/ONNX Runtime) to validate deployment feasibility.
5. **Model compression**: Apply quantization (INT8) and pruning to further reduce model size and latency for deployment on microcontrollers.

5. Conclusions

This paper presents a lightweight deep learning framework integrating MobileNetV3 and EfficientNet-B0 with a Channel-Spatial Attention Module (CSAM) and focal weighted loss. We repurpose FER2013 and CK+ for engagement detection through an emotion-to-engagement mapping, enabling evaluation on publicly available facial expression datasets.

On FER2013, MobileNetV3+CSAM achieves 70.5% accuracy with 1.26M parameters and 0.068G FLOPs, surpassing ResNet-50 (64.0%, 25.6M) by 3.8% while requiring 2.6× less energy per inference. EfficientNet-B0+CSAM reaches 69.1% with 5.4M parameters. The CSAM module contributes 1.3–5.0% accuracy gain with negligible overhead, and the focal weighted loss with weighted sampling improves minority-class F1 by 1.5–2.4% over standard approaches. Cross-dataset evaluation on CK+ (71.9% accuracy without fine-tuning, 114 test samples) provides preliminary evidence of generalizability. Validation on DAiSEE—an engagement-specific dataset with direct annotations—achieves 74.6% validation accuracy and 61.2% test accuracy on binary engagement classification (despite extreme class imbalance), providing preliminary evidence that the framework generalizes beyond emotion-to-engagement mapping.

The computational efficiency of these lightweight models suggests potential for edge deployment, though actual edge hardware validation remains necessary. These findings provide a foundation for developing efficient engagement recognition systems, with the caveat that the emotion-to-engagement mapping requires validation against engagement-specific datasets and that real classroom deployment involves additional challenges (face detection, variable conditions, privacy considerations) not addressed in this study.

Future work will focus on direct evaluation on engagement-specific datasets (e.g., DAiSEE), validation of the emotion-to-engagement mapping against human annotations, temporal modeling with video sequences, actual edge device benchmarking, multimodal fusion, and ethical frameworks for classroom deployment.

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Data Availability Statement: The FER2013 and CK+ datasets used in this study are both publicly available. FER2013 can be accessed via Kaggle (https://www.kaggle.com/datasets/msambare/fer2013). CK+ is available at http://www.consortium.ri.cmu.edu/ckagree/. DAiSEE is available at https://daciiit.github The source code for model implementation is available at https://github.com/SEMHAQ/edu-cv-engagement.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

CNN	Convolutional Neural Network
CSAM	Channel-Spatial Attention Module
FER2013	Facial Expression Recognition 2013
CK+	Extended Cohn-Kanade
FLOPs	Floating Point Operations
MACs	Multiply-Accumulate Operations
SE	Squeeze-and-Excitation
NAS	Neural Architecture Search

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